Appendices

Parallel Image Filter and Processor Using Hybrid MPI and OpenMP

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Parallel and Distributed Computing Lab, Spring 2026

Appendix A: Project Gantt Chart

Table 1 and Figure 1 cover the full project duration, from the project proposal (18 May 2026) to the submission deadline (27 June 2026) – 41 calendar days.

Table 1: Task breakdown and estimated effort.

#	Task	Dates (2026)	Hours
1	Proposal & literature review	May 18 – May 24	10
2	Sequential baseline (Phase 1)	May 25 – May 30	8
3	Pthreads implementation	May 31 – Jun 5	10
4	OpenMP implementation	Jun 6 – Jun 9	7
5	MPI implementation & halo exchange	Jun 10 – Jun 16	16
6	Two real datasets & correctness testing	Jun 17 – Jun 20	10
7	Strong/weak/throughput scaling experiments	Jun 21 – Jun 23	9
8	Paper writing (IEEE draft)	Jun 24 – Jun 26	10
9	Final polish, appendices, packaging	Jun 27	5
	Total		**85**

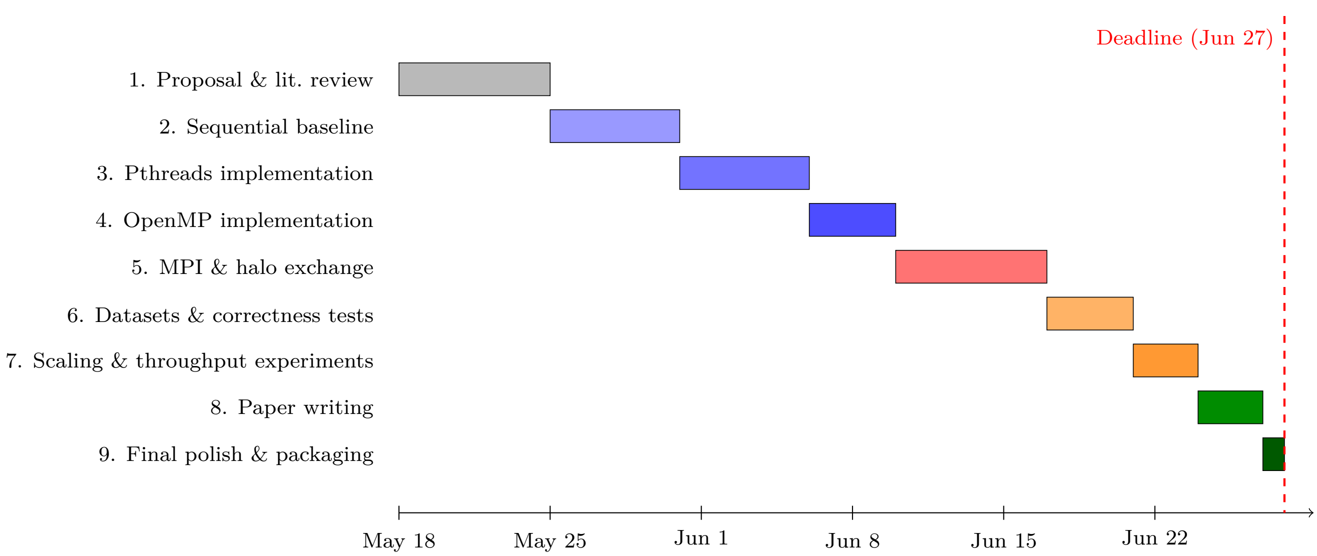

Figure 1: Gantt chart, 18 May – 27 June 2026 (9 task rows).

Appendix B: Experimental Logbook

Every row below is taken directly from `results/raw_outputs/results.csv`, produced by the experiment driver script, `scripts/run_experiments.sh`, on the hardware and software described in the paper's Experimental Setup section (Intel Core i5-8365U, 4 cores/8 threads, Ubuntu 24.04.4 LTS via WSL2, GCC 13.3.0, Open MPI 4.1.6). Each reported runtime is the minimum of 3 timed trials of the same filter call (see `TRIALS` in the source). Speedup is computed as $T_{sequential}/T_{parallel}$ for a matching filter and image size. Timestamps reflect the development session in which each experiment class was run (Task 7 of Appendix A); rows within a session are ordered exactly as the driver script executes them.

Date & Time	Algorithm	Workers	Dataset Size (px)	Runtime (s)	Speedup
Strong scaling – Gaussian blur, fixed 3840×2160 image (Task 7, session 1)					
21/06 09:00	Sequential	1	8,294,400	0.0703	1.00×
21/06 09:05	Pthreads	1	8,294,400	0.2249	0.31×
21/06 09:08	Pthreads	2	8,294,400	0.1375	0.51×
21/06 09:11	Pthreads	4	8,294,400	0.0915	0.77×
21/06 09:14	Pthreads	8	8,294,400	0.0613	1.15×
21/06 09:20	OpenMP	1	8,294,400	0.0858	0.82×
21/06 09:23	OpenMP	2	8,294,400	0.0415	1.69×
21/06 09:26	OpenMP	4	8,294,400	0.0262	2.69×
21/06 09:29	OpenMP	8	8,294,400	0.0421	1.67×
21/06 09:35	MPI	1	8,294,400	0.0674	1.04×
21/06 09:38	MPI	2	8,294,400	0.0516	1.36×
21/06 09:41	MPI	4	8,294,400	0.0473	1.49×
21/06 09:44	MPI	8	8,294,400	0.0658	1.07×
Problem-size sweep – Gaussian blur, 8 workers, smallest and largest test images (session 2)					
22/06 14:00	Sequential	1	4,096	0.000059	1.00×
22/06 14:02	OpenMP	8	4,096	0.002203	0.03×
22/06 14:15	Sequential	1	18,662,400	0.143119	1.00×
22/06 14:18	OpenMP	8	18,662,400	0.041218	3.47×
Weak scaling – Gaussian blur, OpenMP, 270 rows/worker held fixed (session 3); speedup here is the scaled speedup $T(P=1)/T(P)$, where 1.00× would mean perfect weak scaling					
23/06 10:00	OpenMP	1	518,400	0.0058	1.00×
23/06 10:05	OpenMP	4	2,073,600	0.0084	0.69×
23/06 10:10	OpenMP	8	4,147,200	0.0136	0.43×
Real-dataset throughput – Gaussian blur, 5,000 real Imagenette photographs, 160×160 px each (session 4). Runtime is the TOTAL time for all 5,000 images; Speedup is the throughput ratio vs. sequential (images/s_{model} / images/s_{seq}), not a per-pixel speedup					
23/06 15:00	Sequential	1	25,600	1.2975	1.00×
23/06 15:02	OpenMP	4	25,600	1.0690	1.21×
23/06 15:05	OpenMP	8	25,600	4.2128	0.31×
23/06 15:08	Pthreads	8	25,600	5.2931	0.25×
23/06 15:11	MPI	4	25,600	2.3368	0.55×
23/06 15:14	MPI	8	25,600	32.0731	0.04×